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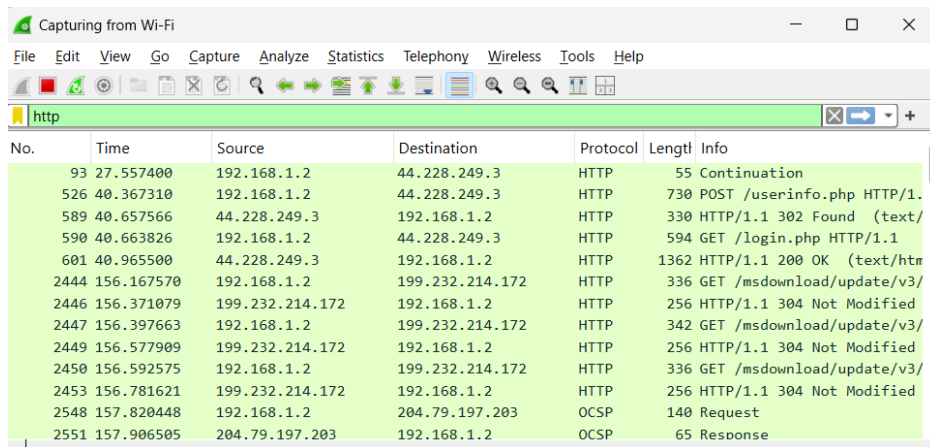
SAP ID:60019220110

SUB: Network Security Lab

EXPERIMENT 1

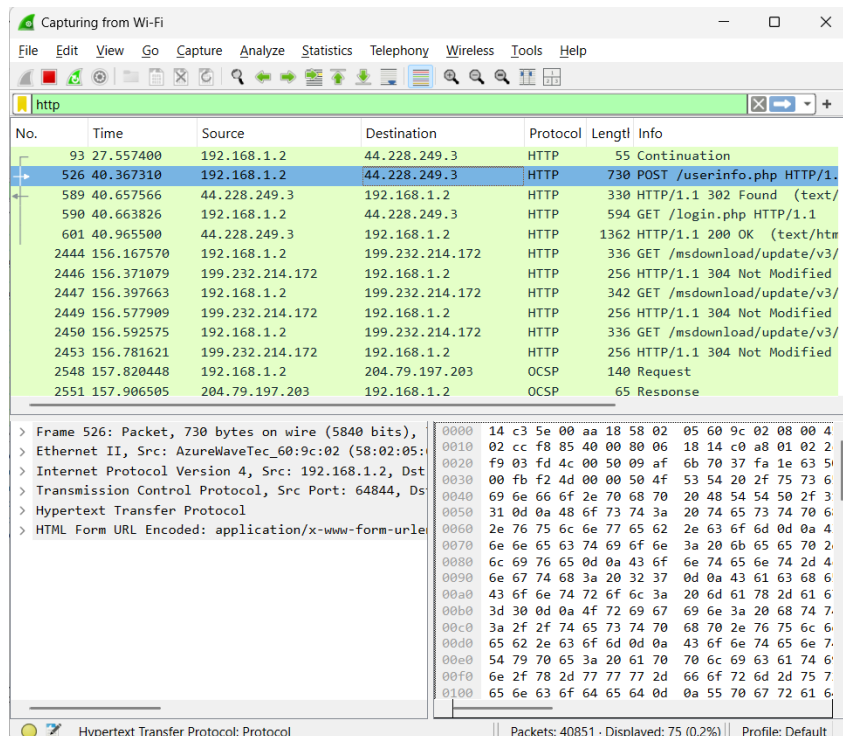
Aim: Study and Implementation of Wireshark Tools

1. Capture HTTP login traffic.



No.	Time	Source	Destination	Protocol	Length	Info
93	27.557400	192.168.1.2	44.228.249.3	HTTP	55	Continuation
526	40.367310	192.168.1.2	44.228.249.3	HTTP	730	POST /userinfo.php HTTP/1.1
589	40.657566	44.228.249.3	192.168.1.2	HTTP	330	HTTP/1.1 302 Found (text/
590	40.663826	192.168.1.2	44.228.249.3	HTTP	594	GET /login.php HTTP/1.1
601	40.965500	44.228.249.3	192.168.1.2	HTTP	1362	HTTP/1.1 200 OK (text/html)
2444	156.167570	192.168.1.2	199.232.214.172	HTTP	336	GET /msdownload/update/v3/
2446	156.371079	199.232.214.172	192.168.1.2	HTTP	256	HTTP/1.1 304 Not Modified
2447	156.397663	192.168.1.2	199.232.214.172	HTTP	342	GET /msdownload/update/v3/
2449	156.577909	199.232.214.172	192.168.1.2	HTTP	256	HTTP/1.1 304 Not Modified
2450	156.592575	192.168.1.2	199.232.214.172	HTTP	336	GET /msdownload/update/v3/
2453	156.781621	199.232.214.172	192.168.1.2	HTTP	256	HTTP/1.1 304 Not Modified
2548	157.820448	192.168.1.2	204.79.197.203	OCSP	140	Request
2551	157.906505	204.79.197.203	192.168.1.2	OCSP	65	Response

2. Identify the POST request.



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> Frame 526: Packet, 730 bytes on wire (5840 bits),
> Ethernet II, Src: AzureWaveTec_60:9c:02 (58:02:05:),
> Internet Protocol Version 4, Src: 192.168.1.2, Dst:
> Transmission Control Protocol, Src Port: 64844, Ds
> Hypertext Transfer Protocol
> HTML Form URL Encoded: application/x-www-form-urle

0000 14 c3 5e 00 aa 18 58 02 05 60 9c 02 08 00 4
0010 02 cc f8 85 40 00 80 06 18 14 c0 a8 01 02 2
0020 f9 03 fd 4c 00 50 09 af 6b 70 37 fa 1e 63 5
0030 00 fb f2 4d 00 00 50 4f 53 54 20 2f 75 73 6
0040 69 6e 66 6f 2e 70 68 70 20 48 54 54 50 2f 3
0050 31 0d 0a 48 6f 73 74 3a 20 74 65 73 74 70 6
0060 2e 76 75 6c 6e 77 65 62 2e 63 6f 6d 0d 0a 4
0070 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2
0080 6c 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 2d 4
0090 6e 67 74 68 3a 20 32 37 0d 0a 43 61 63 68 6
00a0 43 6f 6e 74 72 6f 6c 3a 20 6d 61 78 2d 61 6
00b0 3d 30 0d 0a 4f 72 69 67 69 6e 3a 20 68 74 7
00c0 3a 2f 2f 74 65 73 74 70 68 70 2e 76 75 6c 6
00d0 65 62 2e 63 6f 6d 0d 0a 43 6f 6e 74 65 6e 7
00e0 54 79 70 65 3a 20 61 70 70 6c 69 63 61 74 6
00f0 6e 2f 78 2d 77 77 72 2d 66 6f 72 6d 2d 75 7
0100 65 6e 63 6f 64 65 64 0d 0a 55 70 67 72 61 6

Hypertext Transfer Protocol: Protocol | Packets: 40851 - Displayed: 75 (0.2%) | Profile: Default



3. Extract the username and password from packet details.

```
> Frame 526: Packet, 730 bytes on wire (5840 bits),
> Ethernet II, Src: AzureWaveTec_60:9c:02 (58:02:05:00:02:02), Dst: 08:00:27:00:00:00
> Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.1.1
> Transmission Control Protocol, Src Port: 64844, Dst Port: 80, Seq: 304414412, Win: 0, Len: 0
> Hypertext Transfer Protocol
✓ HTML Form URL Encoded: application/x-www-form-urlencoded
  > Form item: "uname" = "testuser"
  > Form item: "pass" = "test123"
```

4. Write your observation.

- **HTTP Login Traffic Capture**

Wireshark successfully captured HTTP packets when a login request was performed on a website using the HTTP protocol. The packets contained clear-text data because HTTP does not use encryption.

- **Identification of POST Request**

The login request was identified as an HTTP POST request, which is used to send form data from the client to the server. The POST request contained the login credentials in its payload.

- **Extraction of Username and Password**

By expanding the packet details of the POST request, the username and password were visible in plain text under the form data section. This shows that HTTP is insecure for transmitting sensitive information.

- **Source and Destination IP Address**

After applying the filter http, the source IP address was observed as the client machine and the destination IP address was the web server hosting the login page.



1. Apply filter: http & Identify source and destination IP.

Wireshark packet capture showing HTTP traffic. The packet list on the left shows several HTTP GET requests. The packet details pane on the right shows the structure of an HTTP GET request, including the status bar (200 OK) and various headers like Content-Type and Content-Length.

2. Use Follow TCP Stream.

Wireshark packet capture showing a TCP stream. The packet list on the left shows a single TCP packet. The packet details pane on the right shows the structure of a TCP segment, including the sequence number and window size. The packet bytes pane on the right shows the raw data of the TCP segment.



3. View Protocol Hierarchy.

